FURTHER KINEMATICS PROBLEMS IN HIGHER DIMENSIONS **02/03/26**

1. A dog is running round a garden. Its position at time t is given by

$$\mathbf{r} = \begin{pmatrix} \frac{1}{2}t^2 - 7t + 20 \\ \frac{1}{4}t^3 - \frac{17}{4}t^2 + 15t \end{pmatrix}$$

measured in metres from a tree in the centre of the lawn, with the **i** direction aligned due East and the **j** direction due North.

(a) When is the dog due North of the tree?
(b) When is the dog due West of the tree?

2. The position vectors of two particles A and B at time t are given by $\mathbf{r}_{\mathrm{A}} = t\mathbf{i} + \left(t^2 - 2t\right)\mathbf{j}$, and

$\mathbf{r}_{\mathrm{B}} = (t-2)\mathbf{i} + \left(t^2 - 6t + 8\right)\mathbf{j}$

(a) Show that A and B travel along the same path.
(b) What is the difference between the motion of the two particles?

3. The position of a passenger riding on a particular section of the Big Dipper at a fun-fair is given by

$$\mathbf{r} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2t \\ \frac{1}{4}(32 - 6t^2 + t^3) \end{pmatrix}$$

for t ranging from 0 to 5 seconds. The **i**-direction is aligned horizontally, and the **j**-direction vertically upwards.

(a) Find the time he takes to travel from the highest point A to the lowest point B on the path.
(b) Find his velocities at A and at B.
(c) Find his *speeds* at A and at B.

4. Ship A is 5 km due west of ship B and is travelling on a course 035° at a constant but unknown speed v km h^{-1}. Ship B is travelling at a constant 10 km h^{-1} on a bearing of 300°.

a. Write the velocity of each ship as a vector in terms of **i** and **j**, which are unit vectors east and north respectively.
b. Find the position of each ship at time t hours relative to the starting position of ship A.

The ships are on collision course.

c. Find the speed of ship A.
d. How much time elapses before the collision occurs?

5. An owl is initially perched on a tree. It then goes for a short flight which ends when it dives on to a mouse on the ground. The position vector (in metres) of the owl t seconds into its flight is modelled by

$$\mathbf{r} = t^2(6-t)\mathbf{i} + (12.5 + 4.5t^2 - t^3)\mathbf{j}$$

where the foot of the tree is taken to be the origin and the unit vectors **i** and **j** are horizontal and vertical.

a. Draw a graph showing the bird's flight.
b. For how long is the owl in flight?
c. Find the speed of the owl when it catches the mouse and the angle that its flight makes with the horizontal in that instant.
d. Show that the owl's acceleration is never zero during the flight.

6. A particle of mass 15 kg has velocity at time t given by $\mathbf{v} = (4t^3 - 6t)\mathbf{i} + 4\boldsymbol{j} + (2t - 3)\mathbf{k}$. Initially the particle is at the point $(39, -21, 78)$. Point X has coordinates $(9, 19, -42)$.
   a. What is the initial speed of the particle?
   b. What is the initial force acting on the particle? What is the magnitude of this force?
   c. How far is the particle from X initially?
   d. Find the vector displacement of the particle form X at time t.
   e. Does the particle ever reach X?

7. A particle moves with velocity $(3t+5)\mathbf{i}+(2t-2)\mathbf{j}$.

   Its initial position relative to the origin is $-38\mathbf{i}-4\mathbf{j}$

   Find the times at which:

   a. its speed is $\sqrt{212}$
   b. its velocity is in the direction $5\mathbf{i} + 2\mathbf{j}$
   c. its velocity vector makes an angle of $\arctan\left(\frac{8}{17}\right)$ with the horizontal.
   d. the particle is due East of the origin
   e. the bearing of the particle's position, measured from the origin, is $\arctan\left(\frac{81}{38}\right)$.
   f. it is moving directly away from the origin.

8. David has made a space rocket and is heading off for a quick round trip to the moon. His space rocket has an engine which gives the rocket the vertical thrust needed for take off, and a separate thruster system which allows him to steer.

   The rocket takes off and reaches a point in the atmosphere where the effects of air resistance may be ignored. At this point, the rocket is subject to three main forces. Taking the z-axis to measure vertical displacement from the take off point, and the x and y axes to measure horizontal displacement relative to the take off point, the three forces are:

   A) $26000\mathbf{k}\,N$
   B) $-19600\mathbf{k}\,N$
   C) $(2000\mathbf{i}+1500\mathbf{j})\,N$

   i. Identify the cause of each of the forces A, B and C.

   ii. Taking $g = 9.8ms^{-2}$, calculate the mass of the rocket.

   iii. Calculate the rocket's acceleration at this moment.

   Later in the voyage, the rocket's acceleration (in km/h^2) can be expressed in terms of *t*, the time elapsed (in hours) as 6t**i** + (20-3t)**j** +12**k**. This is valid for $2 \le t \le 4$

   a) Given that the rocket's velocity at time 2 is 12**i** + 32**j** find an expression for the rocket's velocity in terms of *t*.

   b) Given that the rocket's position at time 4 is 12**i** + 32**j**, find the rocket's position after 3 hours.

   c) Find the time at which the rocket's velocity is in the direction 9**i** + 11**j** + 4**k**, and the magnitude of the velocity at this time.